Empirical Proof Record: VALIDATED

2026-05-21T23:40:39.192210+00:00

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Proof ID: `dcc46ecf2a2c6031225ce25eae45c7b9…`
Created: 2026-05-21T23:40:39.192210+00:00

1. Claim

> Under sustained load, the substrate stays cognitively alive on real input: (for every cycle that produced a concept set, $\Phi \geq 0.05$). Empty-concept cycles are excluded — with nothing to integrate, Φ=0 is correct, not a defect; their rate is reported separately. The floor is the lowest-Φ real-input cycle across the run.

- **Operationalisation:** Stream 3 passes over 8 substantive stimuli through Kimera's entity target (Takwin); read the per-cycle "phi" and "concepts". Partition cycles into real-input (concepts non-empty) and empty-input. phi_floor = min Φ over real-input cycles; the LTL invariant holds iff that floor $\geq \varphi$_floor. empty_input_rate and the over-all-cycles phi_floor_strict are reported alongside.
- **Threshold:** `phi_floor >= 0.05 phi`
- **H0:** H0: min $\Phi < 0.05$ on real-input cycles — Φ collapses below the floor despite the input producing concepts
- **H1:** H1: min $\Phi \geq 0.05$ on real-input cycles — the substrate stays cognitively alive whenever it has something to integrate

2. Verdict

VALIDATED — observed `0.6155826812407138`
Φ floor 0.6156 over 24 real-input cycles (3 passes $\times$ 8 stimuli); mean Φ 0.6563, band [0.6156, 0.7041], stdev 0.0326; non-collapse rate 100.0%; 0 empty-input cycles excluded (0.0%; strict floor over all cycles 0.6156); 0 cycles produced no Φ; worst real cycle: stimulus 5 @ cycle 21 = 0.6156

3. Evidence

Pillar	Statistic	Value	95% CI	Library
O.flow.phi_stability	phi_floor	0.6155826812407138	—	ophamin 0.82.0

4. Pre-registration

- Registered at: 2026-05-21T23:40:38.828736+00:00
- Config hash: fc0d6fa2b433540d0fc987795a9290da…
- Data hash: 0289f594eb2ea25b582960992cf2db41…

5. Signature

62c0f4fcba981e415e01eea51573172cd3f5a5c8b3497e2f0b334333db8308e6